SemSynth: Combining Semantic Metadata and Synthetic Surrogates for Discovery of Access-Restricted Clinical Datasets

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Motivation Clinical and biomedical datasets are commonly kept behind institutional firewalls due to privacy, consent, and governance. Yet researchers and developers still need to discover datasets, interpret variables, and assess basic quality before requesting access, in line with FAIR expectations [1]. In practice, access requests are often preceded by manual inspection of PDF codebooks or informal spreadsheets. These documents are not easily queryable, typically omit machine-actionable units and code lists, and rarely expose completeness or missingness patterns at a level that supports cross-dataset comparison [2].

Synthetic tabular data is often proposed as a practical surrogate for prototyping, education, and "first look" exploration, but its utility depends on transparency about what is preserved, what is distorted, and what disclosure risks remain [3, 4]. At the same time, semantic metadata for tabular data has matured: recent and foundational work on variable-level modeling argues that columns, variables, code lists, quanities, units, and summary statistics should be first-class resources with explicit semantics and links to controlled vocabularies [2, 5, 6, 7]. However, synthetic data pipelines and semantic publishing pipelines are usually separate: semantics live in local configuration, while synthetic evaluation results are published without a reusable schema.

Demo *SemSynth* is a compact demonstration pipeline that: (1) builds a dataset and variable metadata profile using standard Semantic Web vocabularies, (2) optionally proposes terminology mappings for columns, and (3) generates synthetic surrogates with multiple backends under a shared evaluation harness. The tool is usable as a Python library and command-line interface. It emits uniform, provenance-traceable artifacts (metadata, reports, synthetic CSV/parquet, and metric summaries) and can also build a lightweight DCAT catalog and a minimal UI over generated reports. The intent is not to claim that synthetic data solves access barriers, but that a combined semantic and synthetic "preview bundle" can reduce friction in data discovery and early method development (possibly through FAIR data points [8], see Figure 1).

We focus on two practical research questions: (i) how to represent variable-level semantics, codebooks, and units so that tabular datasets can be queried and compared without row access; and (ii) how to publish synthetic surrogates together with semantic and provenance metadata so that consumers can audit assumptions and evaluation results. Code, documentation, and demo reports are publicly available at https://github.com/bennokr/semsynth.

Implementation Currently, SemSynth discovers and fetches datasets from OpenML and the UCI Machine Learning Repository. Then, each pipeline run may emit dataset metadata (JSON-LD), variable-level metadata exports (JSON-LD and HTML), full-dataset visualizations (UMAP projections [9]), and per-model subdirectories with synthetic tables, metric summaries, and manifests.

Dataset semantics are represented with a variable-level JSON-LD profile based primarily on the DataSet Variable Ontology [2], combined with DCAT, CSVW, SKOS, QUDT/UCUM, and

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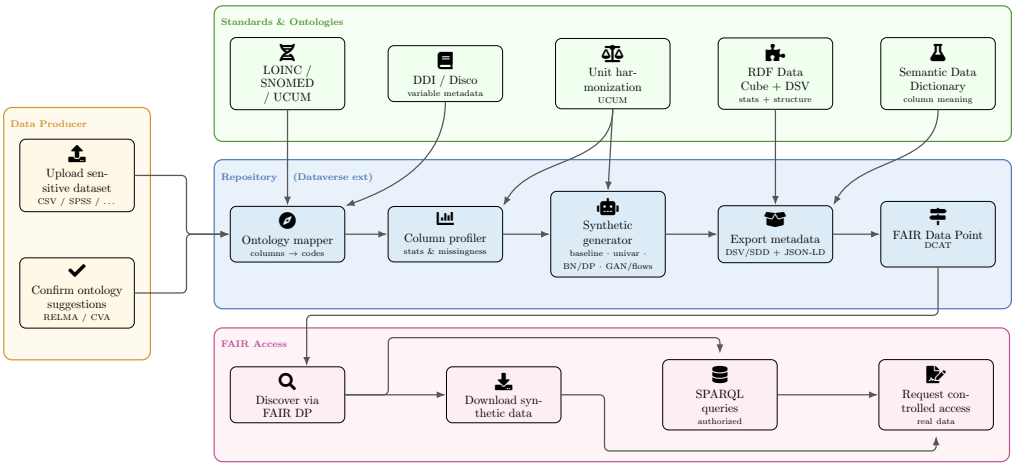

Figure 1: Envisioned possible future SemSynth integration: dataset acquisition and profiling, variable-level metadata export, multi-backend synthesis, and report/provenance emission.

PROV-O. Datasets carry an ordered column schema; columns may include statistical type hints and completeness, units, and codebooks as SKOS concept schemes with external mappings. Metadata can also be embedded in `parquet` files via a `pandas` accessor.

Mapping utilities can suggest terminology codes as SSSOM TSV files [10] using lexical matching, keyword search, embedding re-ranking, and an LLM-assisted workflow [11]; accepted mappings can be merged back into variable metadata or evaluated using benchmarking methods.

SemSynth provides three interchangeable generation backends: MetaSyn [12], PyBNesian [13], and SynthCity [14]. They share data splitting and type inference and write aligned outputs, enabling "apples-to-apples" comparisons in one run. When enabled, SemSynth summarizes disclosure risk (including nearest-neighbor checks, which are popular but flawed [15]) and k-anonymity/t-closeness-style diagnostics on metadata-defined roles [4]. Downstream fidelity compares real vs. synthetic datasets by fitting matched linear predictive models, deriving formulas from metadata, and applying multiple imputation for missing data. SemSynth also tracks provenance via a build graph and emits PROV-O metadata [16].

Artifacts Public demo reports are generated for four well-known health-related datasets from the UCI repository: CDC Diabetes Health Indicators (UCI 891), Chronic Kidney Disease (UCI 336), Heart Disease (UCI 45), and Hepatitis (UCI 46). Each report links the variable-level metadata profile and HTML renderings, summarizes inferred variable types and basic distributions, and lists per-backend fidelity metrics and artifact links.

Limitations The current evaluation is illustrative rather than definitive. First, the demo datasets are public benchmarks and not governed clinical releases; they are best viewed as "health-related" proxies for pipeline behavior. Second, variable-level metadata enrichments are incomplete and partly curated; mapping accuracy and inter-annotator agreement are not yet reported. Third, backends are run with lightweight configurations intended to be broadly runnable, not with tuned hyperparameters or dataset-specific model selection. Fourth, privacy metrics are diagnostic and depend critically on metadata-defined quasi-identifiers and sensitive attributes; they do not provide formal guarantees [15, 4].

Future work Near-term work should focus on (i) systematic evaluation of column-to-terminology mapping quality (including negative results), (ii) explicit modeling of terminology versioning and deprecation, and (iii) repository-facing integration patterns (e.g., ingest-time enrichment,

policy-controlled synthesis, and auditable provenance). Longer-term extensions include multi-table/relational schemas and richer probabilistic metadata beyond current summaries.

Declaration on Generative AI

During the preparation of this work, the author(s) used ChatGPT-5.1 for *Drafting content* and *Peer review simulation*[1]. After using these tool(s)/service(s), the author(s) reviewed and edited the content as needed, fully rewriting most text, and take(s) full responsibility for the publication's content.

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